Use two pistons to move a load

Task 3.4 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch the two syringes, their sizes, the connected tube and the load. Predict travel when the sizes swap. Keep sample and input travel constant.

Measure input and output piston travel with the same reference edge. Record syringe sizes.

Arrangement	Input size	Output size	Load / sample
A			
B swapped			

02 Plan it on paper

Draw the input piston, tube, output piston, and load. Predict the effect of swapping the syringe sizes.

My prediction, chosen feature or design limit

Use two pistons to move a load

Task 3.4 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Mount the connected syringes so an input stroke drives the output piston against a soft sample. Keep hands clear. Swap roles to inspect the mechanism. Run three equal input strokes in both arrangements. Compare output travel and the change in the sample.

Trial	A input / output travel	B input / output travel	Sample change
1			
2			
3			

04 Use the reading

Ch. 19 | Fuel from the Fields | printed pp. 177-184
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which arrangement met the job better? Include travel measurements and identify the trade-off.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.